Exercise 5 (Unstructured Contextual Bandits): Consider a contextual bandit problem with a finite set $\mathcal{X}$ of possible contexts, and a finite set of actions $\mathcal{A}$. Show that running UCB independently for each context yields a regret bound of the order $\widetilde{O}(\sqrt{|X||\mathcal{A}|T})$ in expectation, ignoring logarithmic factors. In the setting where $\mathcal{F} = \mathcal{X} \times \mathcal{A} \to [0,1]$ is unstructured, and consists of all possible functions, this is essentially optimal.

WTS. $\mathrm{Reg} \le \tilde{O}(\sqrt{|X||A|T})$

let x be context and T_x be number of x about whole T rounds.

then this follows $\sum_{x\in\mathcal{X}} T_x = T$

(about std UCB) $\mathrm{Reg}_x \lesssim \tilde{O}(\sqrt{|A|T_x})$ (Prop. 5) about x.

take sum. $\sum_{x\in\mathcal{X}} \mathrm{Reg}_x \lesssim \tilde{O}\left(\sum_{x\in\mathcal{X}}\sqrt{|A|T_x}\right) = \tilde{O}\left(\sqrt{|A|}\cdot\sum_{x\in\mathcal{X}}\sqrt{T_x}\right)$

use Cauchy-schwarz inequality $\sum_{x\in\mathcal{X}} 1\cdot\sqrt{T_x} \le \sqrt{\sum_{x\in\mathcal{X}} 1}\sqrt{\sum_{x\in\mathcal{X}} T_x} = \sqrt{|\mathcal{X}|}\sqrt{\sum_{x\in\mathcal{X}} T_x} = \sqrt{|\mathcal{X}|T}$

thus $\sum_{x\in\mathcal{X}} \mathrm{Reg}_x \lesssim \tilde{O}(\sqrt{|A||\mathcal{X}|T})$ $\longrightarrow$ LinUCB $\mathbf{Reg} \lesssim d\sqrt{T}\log(T)$. (Prop 11)

even if $|\mathcal{X}| \to \infty$, Reg bounded. by dimension d.

Exercise 6 (ε-Greedy with Offline Oracles): In Proposition 8, we analyzed the ε-Greedy contextual bandit algorithm assuming access to an online regression oracle. Because we appeal to online learning, this algorithm was able to handle adversarial contexts $x^1, \dots, x^T$. In the present problem, we will modify the ε-greedy algorithm and proof to show that if contexts are stochastic (that is $x^t \sim \mathcal{D}\ \forall t$, where $\mathcal{D}$ is a fixed distribution), ε-greedy works even if we use an *offline oracle* (Definition 5).

We consider the following variant of ε-greedy. The algorithm proceeds in epochs $m = 0, 1, \dots$ of doubling size

$$\{2\}, \{3,4\}, \{5 \dots 8\}, \dots, \underbrace{\{2^m+1, 2^{m+1}\}}_{\text{epoch } m}, \dots, \{T/2+1, T\};$$

we assume without loss of generality that T is a power of 2, and that an arbitrary decision is made on round $t = 1$. At the end of each epoch $m-1$, the offline oracle is invoked with the data from the epoch, producing an estimated model $\widehat{f}^m$. This model is used for the greedy step in the next epoch m. In other words, for any round $t \in [2^m+1, 2^{m+1}]$ of epoch m, the algorithm observes $x^t \sim \mathcal{D}$, chooses an action $\pi^t \sim \mathrm{unif}[A]$ with probability ε and chooses the greedy action

$$\pi^t = \arg\max_{\pi \in [A]} \widehat{f}^m(x^t, \pi)$$

with probability $1-\varepsilon$. Subsequently, the reward r^t is observed.

1. Prove that for any $T \in \mathbb{N}$ and $\delta > 0$, by setting ε appropriately, this method ensures that with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim A^{1/3}T^{1/3}\left(\sum_{m=1}^{\log_2 T} 2^{m/2}\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, 2^{m-1}, \delta/m^2)^{1/2}\right)^{2/3}$$

for any round $t \in [2^m+1, 2^{m+1}]$, $x^t \sim D$ and $\pi^t \sim \mathrm{unif}[A]$ with ϵ

choose $\pi^t = \arg\max_{\pi \in [A]} \hat{f}^m(x^t, \pi)$ with $1-\epsilon$

① By the ϵ-greedy action,

$$Reg = \mathbb{E}_{x\sim D}[f^*(x,\pi^*) - f^*(x,\pi^t)] \le \epsilon + (1-\epsilon)\mathbb{E}_x[f^*(x,\pi^*) - f^*(x,\hat{\pi}^m)] \quad \text{about } t.$$

here $f^*(x,\pi^*) - f^*(x,\hat{\pi}^m) \le |f^*(x,\pi^*) - \hat{f}^m(x,\pi^*)| + |\hat{f}^m(x,\hat{\pi}^m) - f^*(x,\hat{\pi}^m)|$; triangular inequality.

$$\mathbb{E}_{x\sim D}\left[\left(f^*(x,\pi) - \hat{f}^m(x,\pi)\right)^2\right] \le \frac{A}{\epsilon}\mathbb{E}_{x,a\sim p^{m-1}}\left[\left(f^*(x,a) - \hat{f}^m(x,a)\right)^2\right] \quad \text{about policy } p^{m-1}.$$

by def 5. $\mathbb{E}_{p^{m-1}}\left[(f^* - \hat{f}^m)^2\right] \le \frac{1}{2^{m-1}} Est^{off}_{Sq}\left(F, 2^{m-1}, \frac{\delta}{m^2}\right)$,

by offline oracle (m-1) epoches.

ϵ-greedy. Algorithm. $p^{m-1}(a|x) \ge \frac{\epsilon}{A}$

Sum. $[1,T]$

$$Reg \le \epsilon T + \sum_{m=1}^{\log_2 T} 2^m \cdot 2\sqrt{\frac{A}{\epsilon}}\left(\frac{1}{2^{m-1}}Est^{off}_{Sq}\right)^{\frac{1}{2}} = \epsilon T + 2\sqrt{2}\sqrt{\frac{A}{\epsilon}}\underbrace{\sum_{m=1}^{\log_2 T} 2^{\frac{m}{2}}(Est^{off}_{Sq})^{\frac{1}{2}}}_{\text{let } S} =: f$$

$$\frac{\partial f}{\partial \epsilon} = T + 2\sqrt{2}\cdot\frac{1}{2}\left(\frac{A}{\epsilon}\right)^{-\frac{1}{2}}\cdot\left(\frac{-A}{\epsilon^2}\right)S = 0, \quad \frac{T}{\sqrt{2}S} = \frac{A^{\frac{1}{2}}}{\epsilon^{3/2}}$$

$$\text{take. } \epsilon = \left(\frac{2^{\frac{1}{2}}A^{\frac{1}{2}}S}{T}\right)^{\frac{2}{3}} = \left(\frac{2AS^2}{T^2}\right)^{\frac{1}{3}}$$

$$\text{thus, } Reg \lesssim \epsilon T + 2\sqrt{2}S\sqrt{\frac{A}{\epsilon}} = \left(\frac{2AS^2}{T^2}\right)^{\frac{1}{3}}\cdot T + 2\sqrt{2}\cdot S A^{\frac{1}{2}}\cdot\left(\frac{\sqrt{2}A^{\frac{1}{2}}S}{T}\right)^{-\frac{1}{3}}$$

$A^{\frac{1}{3}}$

$$= 2^{\frac{1}{3}}A^{\frac{1}{3}}S^{\frac{2}{3}}T^{\frac{1}{3}} + 2\sqrt{2}S A^{\frac{1}{3}}S^{\frac{2}{3}}T^{\frac{1}{3}}$$

$$= (2^{\frac{1}{3}} + 2\sqrt{2})A^{\frac{1}{3}}S^{\frac{2}{3}}T^{\frac{1}{3}}$$

$$= O(1)(AT)^{\frac{1}{3}}\left(\sum_{m=1}^{\log_2 T} 2^{\frac{m}{2}}(Est^{off}_{Sq})^{\frac{1}{2}}\right)^{\frac{2}{3}}$$

2. Recall that for a finite class, ERM achieves $\mathbf{Est}^{\mathsf{off}}_{\mathsf{Sq}}(\mathcal{F}, T, \delta) \lesssim \log(|\mathcal{F}|/\delta)$. Show that with this choice, the above upper bound matches that in Proposition 8, up to logarithmic in T factors.

- $Est^{off}_{Sq}(F, T, \delta) \lesssim \log(|F|/\delta)$

$$S \approx \sqrt{\log(|F|/\delta)}\sum_{m=1}^{\log_2 T}(\sqrt{2})^m \approx O\left(\sqrt{T\log(|F|/\delta)}\right)$$

dominant.

$$Reg \le O(1)(AT)^{\frac{1}{3}}S^{\frac{2}{3}} = O(1)\left(AT^2\log(|F|/\delta)\right)^{\frac{1}{3}}$$

Exercise 17

1. Show that for any sequence of estimators $\widehat{f}^1, \ldots, \widehat{f}^t$, by choosing $p^t = \mathsf{IGW}_\gamma(\widehat{f}^t(x^t, \cdot))$, we have that

$$\mathbf{Reg} = \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}[f^\star(x^t, \pi^\star(x^t)) - f^\star(x^t, \pi^t)] \lesssim \frac{AT}{\gamma} + \gamma \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] + \varepsilon T.$$

If we had $f^\star = \bar{f}$, this would follow from Proposition 9, but the difference is that in general ($\bar{f} \neq f^\star$), the expression above measures estimation error with respect to the best-in-class model $\bar{f}$ rather than the true model $f^\star$ (at the cost of an extra εT factor).

from ϵ-misspecified. condition

$|f^*(x,a) - \bar{f}(x,a)| \le \epsilon$ any action a. $\Rightarrow$ $f^*(x,a) \le \bar{f}(x,a) + \epsilon$ and $-f^*(x,a) \le -\bar{f}(x,a) + \epsilon$

$$\begin{cases} f^*(x,\pi^*) \le \bar{f}(x,\pi^*) + \epsilon & \text{for } \pi^* \\ -f^*(x,\pi^t) \le -\bar{f}(x,\pi^t) + \epsilon & \text{for } \pi^t \end{cases}$$

$$\Rightarrow f^*(x,\pi^*) - f^*(x,\pi^t) \le \bar{f}(x,\pi^*) - \bar{f}(x,\pi^t) + 2\epsilon$$

$$\le \Big(\bar{f}(x, \underbrace{\pi^*}_{\text{optimal action about } \bar{f}}) - \bar{f}(x,\pi^t)\Big) + 2\epsilon$$

take p^t, Prop 9. then $\mathbb{E}_{\pi^t \sim p^t}\left[\bar{f}(x^t,\pi^*) - \bar{f}(x^t,\pi^t)\right] \le \frac{A}{\gamma} + \gamma\, \mathbb{E}_{\pi^t \sim p^t}\left[\left(\hat{f}^t(x^t,\pi^t) - \bar{f}(x^t,\pi^t)\right)^2\right]$

$$\mathrm{Reg} \le \sum_{t=1}^{T}\left(\frac{A}{\gamma} + \gamma\, \mathbb{E}_{\pi^t \sim p^t}\left[\left(\hat{f}^t(x^t,\pi^t) - \bar{f}(x^t,\pi^t)\right)^2\right] + 2\epsilon\right)$$

$$\simeq \frac{AT}{\gamma} + \gamma \sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[\left(\hat{f}^t(x^t,\pi^t) - \bar{f}(x^t,\pi^t)\right)^2\right] + 2\epsilon T.$$

2. Show that the following inequality holds for every sequence

$$\sum_{t=1}^{T} (\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 \le \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F}, T) + 2\sum_{t=1}^{T} (r^t - \bar{f}(x^t, \pi^t))(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t)).$$

Since $\sum_{t=1}^{T}\left(\hat{f}^t(x^t,\pi^t) - r^t\right)^2 - \sum_{t=1}^{T}\cancel{\left(\bar{f}(x^t,\pi^t) - r^t\right)^2} \le \mathrm{Reg}_{Sq}(\mathcal{F},T)$

here, $\left(\hat{f}^t - \bar{f} + \bar{f} - r^t\right)^2 = \left(\hat{f}^t - \bar{f}\right)^2 + 2\left(\hat{f}^t - \bar{f}\right)\left(\bar{f} - r^t\right) + \cancel{\left(\bar{f} - r^t\right)^2}$

$$\mathrm{LHS} = \sum_{t=1}^{T}\left(\hat{f}^t - \bar{f}\right)^2 + 2\left(\hat{f}^t - \bar{f}\right)\left(\bar{f} - r^t\right) \le \mathrm{Reg}_{Sq}(\mathcal{F},T)$$

$$\sum_{t=1}^{T}\left(\hat{f}^t - \bar{f}\right)^2 \le \mathrm{Reg}_{Sq}(\mathcal{F},T) + 2\sum_{t=1}^{T}\left(\hat{f}^t - \bar{f}\right)\left(r^t - \bar{f}\right)$$

3. Using Freedman's inequality (Lemma 35), show that with probability at least $1 - \delta$,

$$\sum_{t=1}^{T} \mathbb{E}_{\pi^t \sim p^t}\left[(\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2\right] \le 2\sum_{t=1}^{T} (\widehat{f}^t(x^t, \pi^t) - \bar{f}(x^t, \pi^t))^2 + O(\log(1/\delta)).$$

Lemma 35 (Freedman's inequality (Bernstein for martingales)): Let $(X_t)_{t \le T}$ be a real-valued martingale difference sequence adapted to a filtration $(\mathscr{F}_t)_{t \le T}$. If $|X_t| \le R$ almost surely, then for any $\eta \in (0, 1/R)$, with probability at least $1 - \delta$, for all $T' \le T$,

$$\sum_{t=1}^{T'} X_t \le \eta \sum_{t=1}^{T'} \mathbb{E}_{t-1}[X_t^2] + \frac{\log(\delta^{-1})}{\eta}.$$

Since $\hat{f}^t, \bar{f}^t \in [0,1]$ also their difference in $[0,1]$

let $Y_t = \left(\hat{f}^t - \bar{f}^t\right)^2 \le 1$ and $z_t = \mathbb{E}(Y_t \mid \mathcal{F}_{t-1}) - Y_t \le 1$

then using Freedman's inequality.

$$\sum_{t=1}^{T} z_t \le h \sum_{t=1}^{T} \mathbb{E}_{t-1}\left[z_t^2\right] + \frac{\log(\delta^{-1})}{h}$$

$$\sum_{t=1}^{T}\left(\mathbb{E}(Y_t \mid \mathcal{F}_{t-1}) - Y_t\right) \le h\sum_{t=1}^{T} \mathbb{E}(Y_t \mid \mathcal{F}_{t-1}) + \frac{\log(\delta^{-1})}{h}$$

$\because \mathrm{Var}(z_t \mid \mathcal{F}_{t-1}) \le \mathbb{E}(Y_t \mid \mathcal{F}_{t-1})$ where $Y_t^2 \le Y_t$

take $h = \frac{1}{2}$, then $\sum_{t=1}^{T} \mathbb{E}(Y_t \mid \mathcal{F}_{t-1}) \le 2\sum_{t=1}^{T} Y_t + \log(\delta^{-1})$

4. Using Freedman's inequality once more, show that with probability at least $1-\delta$,

$$2\sum_{t=1}^{T}(r^t-\bar{f}(x^t,\pi^t))(\widehat{f}^t(x^t,\pi^t)-\bar{f}(x^t,\pi^t)) \le \frac{1}{4}\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[(\widehat{f}^t(x^t,\pi^t)-\bar{f}(x^t,\pi^t))^2\right]+O(\varepsilon^2T+\log(1/\delta)).$$

Conclude that with probability at least $1-\delta$,

$$\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[(\widehat{f}^t(x^t,\pi^t)-\bar{f}(x^t,\pi^t))^2\right] \lesssim \mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F},T)+\varepsilon^2T+\log(1/\delta).$$

Since $r^t-\bar{f}^t = \underbrace{r^t-f^{*t}}_{\text{통계적 노이즈. (A)}} + \underbrace{f^{*t}-\bar{f}^t}_{\text{bound로 발생. (B)}}$

about (B) $\quad 2(f^{*t}-\bar{f}^t)(\hat{f}^t-\bar{f}^t) \le \frac{1}{4}(\hat{f}^t-\bar{f}^t)^2+4(f^{*t}-\bar{f}^t)^2 \quad$ (Peter-Paul inequality, $\epsilon=\frac{1}{4}$)

$$\le \frac{1}{4}(\hat{f}^t-\bar{f}^t)^2+4\epsilon^2$$

ϵ-misspecified condition, $f^{*t}-\bar{f}^t \le \epsilon^2$

$$\sum_{t=1}^{T} \text{LHS} \le \sum_{t=1}^{T}\frac{1}{4}(\hat{f}^t-\bar{f}^t)^2+4\epsilon^2T \quad \therefore$$

(A) let martingales $M_t = 2(r^t-f^{*t})(\hat{f}^t-\bar{f}^t)$

use freedman inequality, $\eta=\frac{1}{16}$

$$\sum_{t=1}^{T}M_t \le \frac{1}{4}\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[(\hat{f}^t-\bar{f}^t)^2\right]+16\log(\delta^{-1}) \quad \therefore$$

done by (A)+(B)

5. Combining the previous results, show that for any $\delta>0$, by choosing $\gamma>0$ appropriately, we have that with probability at least $1-\delta$,

$$\mathbf{Reg} \lesssim \sqrt{AT\cdot(\mathbf{Reg}_{\mathsf{Sq}}(\mathcal{F},T)+\log(1/\delta))}+\varepsilon\cdot A^{1/2}T.$$

Since 1. $\mathrm{Reg} \le \frac{AT}{\gamma}+\gamma\sum_{t=1}^{T}\mathbb{E}_{\pi^t\sim p^t}\left[(\hat{f}^t-\bar{f}^t)^2\right]+\epsilon T \quad \leftarrow \sum_{t=1}^{T}\mathbb{E}(\hat{f}^t-\bar{f}^t)^2 \le \mathrm{Reg}_{sq}+\epsilon^2T+\log(\frac{1}{\delta})$

$$\le \frac{AT}{\gamma}+\gamma\left(\mathrm{Reg}_{sq}+\epsilon^2T+\log(\tfrac{1}{\delta})\right)+\epsilon T$$

let $C_1 = AT$, $\quad C_2 = \mathrm{Reg}_{sq}+\epsilon^2T+\log\frac{1}{\delta}$

$\frac{C_1}{\gamma}+C_2\gamma \ge 2\sqrt{C_1C_2}$ (산술기하 등호조건) $\quad \frac{C_1}{\gamma_{min}} = C_2\gamma_{min} \Leftrightarrow \gamma_{min} = \sqrt{\frac{C_1}{C_2}}.$

thus $\mathrm{Reg} \le C_1\sqrt{\frac{C_2}{C_1}}+\sqrt{\frac{C_1}{C_2}}\cdot C_2+\epsilon T = 2\sqrt{C_1C_2}+\epsilon T$

$$= O\left(\sqrt{AT(\mathrm{Reg}_{sq}+\epsilon^2T+\log(\delta^{-1}))}\right)+\epsilon T$$

Since $\sqrt{AT(\mathrm{Reg}_{sq}+\log(\delta^{-1}))+\epsilon^2T} \le \sqrt{AT(\mathrm{Reg}_{sq}+\log(\delta^{-1}))}+\sqrt{A\epsilon^2T}$

$$\mathrm{RHS} \le O\left(\sqrt{AT(\mathrm{Reg}_{sq}+\log(\delta^{-1}))}\right)+\epsilon\underbrace{(A^{\frac{1}{2}}+1)}_{A\text{ dominant.}}T \approx O\left(\sqrt{AT(\mathrm{Reg}_{sq}+\log(\delta^{-1}))}\right)+\epsilon A^{\frac{1}{2}}T$$